

Grey goo as a technosignature: propagation timescales from planet to cosmos, the observational record, and what its silence does and does not say about our own risk

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Abstract

“Grey goo”—a self-replicating machine population that has escaped every control on its growth—has been discussed for forty years as a planetary catastrophe and, separately, self-replicating probes have been discussed for fifty as a way of exploring the Galaxy. The two literatures have not met: no published work asks how far an *uncontrolled* replicator propagates, on what timescale, what it would look like from outside, and what the fact that we do not see it implies for the probability that a technological civilisation destroys itself this way. We supply the missing calculation. We separate the question by scale and by the one property that governs it, dispersal. In place on a planet, replication is never the bottleneck: a biosphere is consumed in hours to days at any plausible doubling time, but a crust takes ~ 4.5 kyr and a bulk planet ~ 1.1 Myr because the surface cannot shed the heat of processing faster than $4\pi R^2 \sigma T^4$ at the temperature the machines survive. In a planetary system the small bodies are transport-limited (14 yr for the asteroid belt) and produce a Dyson-scale waste-heat source at the feedstock’s orbital temperature (169 K for the belt); we show that a *detectable* dead residue is necessarily a *short-lived* one, surviving at most $P_{\text{orb}}/f_{\text{min}}$ above the detection floor f_{min} (4.4 kyr at 2.7 AU for $f_{\text{min}} = 10^{-3}$), so that the existing waste-heat surveys constrain live goo and barely constrain dead goo. Between stars there are two roads. A replicator never designed to travel still leaves its system by radiation-pressure blowout if it is smaller than $0.57 \mu\text{m}$, and the same radiation pressure lets only the $0.21\text{--}0.57 \mu\text{m}$ size class reach the habitable zone of the next star; a single asteroid belt’s worth of such machines, phase-mixed around the solar annulus in ~ 1.2 Gyr, delivers of order 200 intact devices per year to every rocky planet in the annulus before survival losses, so that if a machine can stay functional in the interstellar medium for $\gtrsim 10^8$ yr the passive branch seeds a galactic annulus in a gigayear without any spacecraft. The active branch—a replicator that inherits interstellar propulsion—fills the Galaxy in 639 kyr–504 Myr and reaches $7 \times 10^8\text{--}5 \times 10^9$ galaxies before the cosmic event horizon closes at 16.6 Gly. We compile what existing searches say at each scale and show that the constraints separate cleanly into three tiers: our own system’s intact belt and living biosphere (anthropically shadowed), the $\sim 5 \times 10^6$ -star waste-heat surveys (unshadowed but short-memoried), and the 10^5 -galaxy $\hat{G}$ survey (unshadowed and long-memoried). A Bayesian treatment with and without the anthropic shadow gives, per civilisation, $p \lesssim 3.0 \times 10^{-7}$ for a persistent galaxy-spanning goo if $\Lambda \sim 100$ civilisations have arisen per galaxy, but *no* constraint on the planet-bound branch at any Λ . Mapped back onto an elicited planet-scale accident probability, the astronomical record moves it by a factor 0.990 if one per cent of such accidents escape their system, and by at most 0.50 if half do. The silence is informative about goo that travels and silent about goo that stays home; the goo we can accidentally make is, for the next several centuries, the kind that stays home. We state what observation would change this and generalise the argument to the other ways a civilisation can die of its own replicators.

1 Introduction

The phrase “grey goo” was coined by Drexler [1986] for a thought experiment: a molecular assembler that copies itself from ambient matter, escapes whatever was meant to stop it, and converts the biosphere into more of itself. The scenario acquired quantitative form in Freitas [2000], who bounded the rate of “global ecophagy” by the heat the replicators must shed, and was then disowned by its author on engineering grounds [Phoenix and Drexler, 2004]: self-replication is not needed for manufacturing, and a replicator able to eat an unprepared environment is a much harder thing to build than one that runs in a vat. The scenario nonetheless remains a standing entry in the existential-risk literature [Bostrom, 2002, Rees, 2003, Ord, 2020], and every survey of that literature assigns it a small but non-zero probability.

A separate literature, older and larger, treats self-replicating machines in space as a good idea. Tipler [1980] argued from von Neumann probes that a single civilisation could occupy the Galaxy in $\lesssim 3 \times 10^8$ yr and that the absence of such probes here shows there is no one else; Freitas [1980] designed one; Hart [1975], Newman and Sagan [1981], Landis [1998], Bjørk [2007], Cotta and Morales [2009], Wiley [2011], Nicholson and Forgan [2013] and Carroll-Nellenback et al. [2019] computed the colonisation front in increasing detail; Armstrong and Sandberg [2013] and Olson [2015] carried it to intergalactic scales, and Hanson et al. [2021] built a cosmological model of “grabby” expansion on it. Within that literature the one paper that reads like a goo argument is Sagan and Newman [1983], who held that a self-reproducing probe population would become a danger to its own makers and would therefore be suppressed by any prudent civilisation, a position Tipler rejected.

What neither literature contains is the calculation the title of this paper names. The planetary goo literature stops at the planet; the probe literature assumes design, purpose and control. The nearest approaches are Stevens et al. [2016], who list grey goo among the ways a civilisation might destroy itself and sketch its planetary signature (a dead, chemically altered surface, perhaps with an artificial spectral feature), and the recent papers on self-replicating probes as solar-system technosignatures [Ellery, 2022a,b, 2025], which are concerned with designed probes and their curbing. Nobody has asked how far an *uncontrolled* replicator can go, how long it takes to get there, what it would look like at each scale, whether the data already exclude it, and—the question with a practical stake—whether the answer should move our estimate of the probability that *we* do it.

We answer these questions in order. Section 2 fixes what we mean by goo and identifies dispersal, not replication rate, as the property that sets its reach. Sections 3–6 compute the timescales at the planetary, system, galactic-passive, galactic-active and cosmological scales, and in each case say what the residue looks like and how long it lasts. Section 7 compiles the observational record. Section 8 does the inference, with and without the anthropic shadow, and maps it onto our own risk. Section 9 generalises to the other self-replicating technologies with which a civilisation could kill itself, and Section 10 states the assumptions on which the argument turns and the observations that would change it. Every number in the paper is generated by the released code from a single parameter file, and where a parameter is uncertain by orders of magnitude we scan it rather than choose it.

2 What grey goo is, physically

We take a replicator to be any system that builds functional copies of itself from its surroundings. It is characterised by four numbers: the doubling time τ under favourable conditions; the energy e_{rep} consumed per unit mass of replicator built; the range of feedstock it can use, which we do not model

but which enters every result as an assumption of *generality*; and the mass m of the unit, which through its size s sets how it moves under gravity and light. We use $\tau = 10^3$ s, $e_{\text{rep}} = 1 \times 10^7$ J kg $^{-1}$ (one electron-volt per atom placed), $s = 0.5$ μm and $\rho = 2$ g cm $^{-3}$ as fiducial, and scan τ from 100 s to a year. The fiducial device has mass 1.0×10^{-15} kg; from one such seed a planet’s mass is 132 doublings away.

“Goo” is a replicator population without an external control on its growth. It is not defined by malevolence, purpose or intelligence; a goo can be a manufacturing organism, a terraforming seed, an industrial swarm or a probe whose replication limiter failed. What it is defined by is that it keeps going until something physical stops it. Four things can: exhaustion of feedstock it can use, the heat of its own metabolism, the error catastrophe that afflicts any finite-fidelity copier [Eigen, 1971, Error-catastrophe authors, 2016], and predation or competition by other replicators [Lotka–Volterra probe authors, 2022]. We treat the first two quantitatively; the last two enter as a residue-lifetime parameter and as a caveat.

The property that determines how far a goo goes is none of these. It is *dispersal*: whether the units can leave the body they are on, the system they are in, and the galaxy. A goo that cannot leave its planet is a planetary catastrophe with no astronomical consequence; a goo that has space transport is a system-scale event; a goo whose units are small enough to be blown out of the system by starlight is an interstellar event on a gigayear timescale; a goo that inherited interstellar propulsion is a galactic event on a megayear timescale. Replication rate sets how fast each scale is eaten once reached, and we will find that it is almost never the bottleneck; dispersal sets which scales are reached at all. We therefore organise the paper by dispersal class, and the inference in Section 8 will turn on the fraction of goo accidents that fall into each.

3 Planet scale: the three ceilings

The number usually quoted for goo is the exponential one: $\log_2(M/m)$ doublings at τ each. For Earth’s biosphere (1.1×10^{15} kg dry; Bar-On et al. 2018) that is 100 doublings and 28 h at $\tau = 10^3$ s; 2.8 h at 100 s, 101 yr at a year. For the crust it is 124 doublings and for the bulk planet 132; the extra 30 doublings cost a day. This is the source of the folk belief that goo eats worlds in days, and it is wrong beyond the biosphere for a reason Freitas [2000] identified for the biosphere itself: heat.

Every kilogram of replicator built dissipates e_{rep} as heat, whatever the energy source—sunlight, the free energy of the feedstock, or fission. A planetary surface at temperature T can shed at most $4\pi R^2\sigma(T^4 - T_{\text{eq}}^4)$, which for Earth is 1.7×10^{18} W at 500 K and 4.6×10^{20} W at 2000 K. A machine that must stay below 500 K to function therefore cannot process more than 1.7×10^{11} kg s $^{-1}$ planet-wide, and the crust takes 4.5 kyr, the bulk 1.1 Myr, at any τ . This radiative ceiling is independent of where the energy comes from; the second ceiling is where it comes from. Absorbed sunlight is 1.2×10^{17} W, which alone would stretch the crust to 62 kyr and the bulk to 16 Myr; the crust’s uranium and thorium hold 8×10^{30} J and the oceans’ deuterium 7×10^{31} J, together enough for the crust and marginally for the bulk, so that a goo with nuclear metabolism is radiatively rather than solar-limited. The governing timescale is the slowest of the three (Table 1): hours for the biosphere, 4.5 kyr for the crust, 1.1 Myr for the planet, the last two independent of τ from 100 s to a year.

Two remarks. First, these are conversions in place; *lifting* the mass off the planet is a different budget, set by the gravitational binding energy $3GM^2/5R$: a week at the full solar luminosity (7 d, Dyson’s famous number), 15 kyr at the 2000 K surface ceiling, 58 Myr on absorbed sunlight alone. A goo confined to a planet does not dismantle it into a swarm; it converts it into a planet made of goo. Second, Freitas [2000] reached a similar conclusion for the biosphere by a different

Table 1: In-place conversion of Earth’s reservoirs at $\tau = 10^3$ s, $e_{\text{rep}} = 1 \times 10^7 \text{ J kg}^{-1}$, machines limited to 500 K. The governing time is the slowest ceiling. Raising the operating temperature to 2000 K shortens the crust to 16 yr and the bulk to 4.1 kyr, which is why a goo that can run hot is a qualitatively worse thing.

reservoir	doublings	exponential	radiative (500 K)	solar-powered
biosphere ($1.1 \times 10^{15} \text{ kg}$)	100	28 h	1.8 h	25 h
crust ($2.4 \times 10^{22} \text{ kg}$)	124	$\sim 1.5 \text{ d}$	4.5 kyr	62 kyr
bulk ($6.0 \times 10^{24} \text{ kg}$)	132	$\sim 1.5 \text{ d}$	1.1 Myr	16 Myr

route—he bounded the global replicator power at $\sim 10^{13} \text{ W}$ by the warming it would cause and found ecophagy times of order a year or more at that limit, and argued that the thermal signature of a replicating patch would be detectable from orbit well before it was global. Our ceilings are the physical ones a goo indifferent to the climate would hit; his is the one a defender could use.

What it looks like from outside. A planet during ecophagy re-radiates the energy it always did plus the metabolism of the goo, which at the radiative ceiling is a $\sim 500 \text{ K}$ surface—a hot rocky planet with the wrong equilibrium temperature for its orbit. Afterwards it is a dead world with a refined surface: metallic or graphitic, uniform, without the disequilibrium chemistry of a biosphere [Stevens et al., 2016]. Nothing in the current exoplanet census can see either. Thermal emission from rocky planets has been measured for a handful of hot, close-in worlds [Kreidberg et al., 2019, Greene et al., 2023, Zieba et al., 2023] and reflects bare rock; surface composition of a temperate planet is beyond any funded facility. The planetary branch is therefore unobservable by construction for the next decades, and this will matter in Section 8.

4 System scale: the belt, the planets, and the residue

A goo with any space transport—it ate a spacefaring civilisation’s infrastructure, or was one—faces a different problem: the feedstock is in 10^6 bodies spread through a torus. Inside a belt the front is a branching process. Each converted body launches b packets to its neighbours; a hop covers the mean spacing at a Δv of a kilometre per second and waits an orbital period for phasing; a body saturates after ~ 100 local doublings. With $b = 100$ the 10^6 -body main belt is consumed in three generations, 14 yr, and the 10^9 -body Kuiper belt in 1.1 kyr—limited by transport and orbital phasing, not by τ , which it hardly sees. Planets again cost more: their mass leaves at the rate the surface can shed the heat, 7.7 kyr for a terrestrial planet and 143 kyr for a giant at the 2000 K ceiling (52 h and 21 yr if the full stellar luminosity could be delivered to the job, which it cannot). The system is therefore eaten from the outside in, small bodies first, and Figure 1 shows the whole ladder.

The residue. The converted mass is a swarm. Spread at areal density Σ on a shell at radius r it covers a fraction $f = \min(1, M/4\pi r^2 \Sigma)$ of the star; the main belt’s $2.4 \times 10^{21} \text{ kg}$ covers the star completely as micron-thick devices ($\Sigma = 10^{-3} \text{ kg m}^{-2}$), a fraction 0.0012 as kilogram-per-square-metre sheets, and 1.2×10^{-6} as metre-scale slabs. A thin absorber at r radiates at $T = 278 \text{ K} (r/\text{AU})^{-1/2}$: 169 K for the belt, 44 K for the Kuiper belt. A live system-scale goo is therefore a partial Dyson sphere at the temperature of its feedstock’s orbit, and every waste-heat search ever run [Carrigan, 2009, Wright et al., 2014b,a, Suazo et al., 2022, 2024] is a search for it,

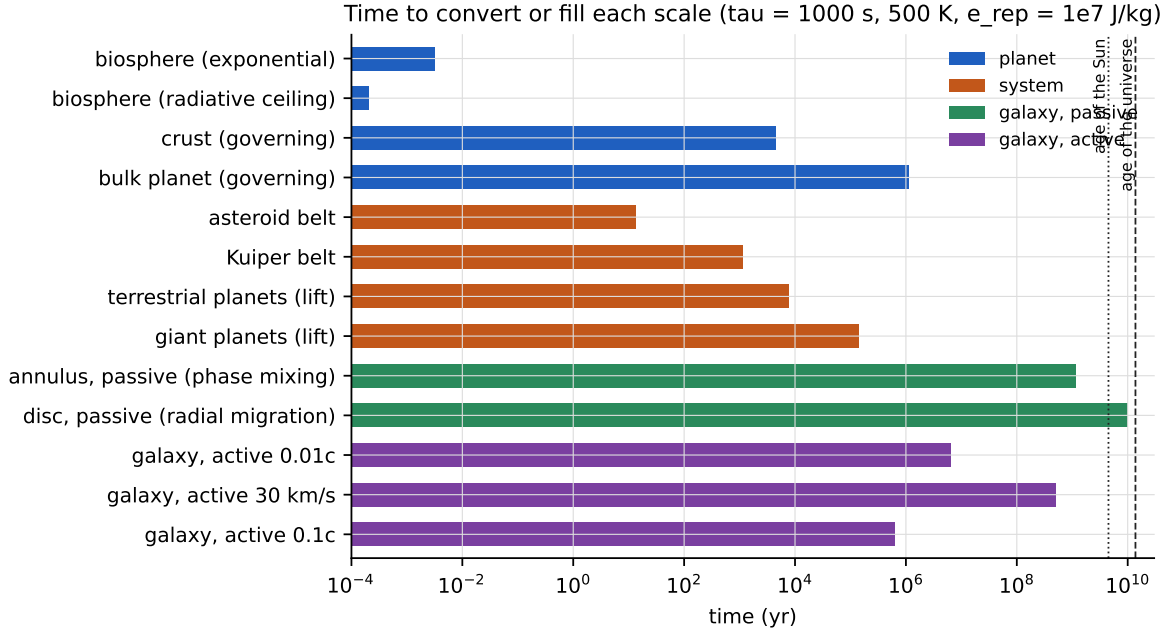


Figure 1: Time to convert or fill each scale, by dispersal class. Planetary conversion is set by the radiative ceiling; system-scale conversion by transport; the passive interstellar branch by galactic phase mixing; the active branch by ship speed.

with one difference in expectation: a purposeful swarm sits where its builders want the energy, a goo sits where the rock was.

A dead one does not last. Lacki [2025] showed that an unmaintained swarm of covering fraction f grinds itself down on the collisional time $\sim P_{\text{orb}}/f$; the fragments then leave by Poynting–Robertson drag [Burns et al., 1979] on 400 yr $(r/\text{AU})^2/\beta$ or, below the blowout size, on one orbit. For the belt mass as 0.5 μm machines the residue is gone in 4.4 yr; as millimetre grains ($f = 4.4 \times 10^{-4}$) in 10 kyr; as metre slabs ($f = 4.4 \times 10^{-7}$) it lasts 10 Myr but is far below any detection floor. This is a general statement, and it is the one that governs what the surveys can say: a residue detectable as an infrared excess needs $f \geq f_{\text{min}}$, and its collisional lifetime is then at most

$$t_{\text{vis}} \leq \frac{P_{\text{orb}}}{f_{\text{min}}}, \quad (1)$$

which at 2.7 AU is 4.4 kyr for $f_{\text{min}} = 10^{-3}$, 444 yr for 10^{-2} and 44 yr for 10^{-1} (25 kyr at 40 AU for 10^{-2}). *A detectable dead residue is a short-lived one.* Figure 2 draws the consequence. The only long-lived system-scale signature is a goo that is still alive—still replicating on its own collision fragments, still re-radiating its captured starlight—and that is what the Dyson searches bound.

5 Between stars, passively: goo as interstellar dust

A replicator never designed to travel has one road out of its system, and it is the road interstellar dust takes. Around the Sun the ratio of radiation pressure to gravity on a sphere is $\beta = 0.57 Q_{\text{pr}}/(\rho s)$ with ρ in g cm^{-3} and s in microns [Burns et al., 1979]; devices smaller than 0.57 μm at $\rho = 2$ have $\beta > 0.5$ and are unbound the moment they are released from an orbiting body, leaving on a hyperbola with $v_{\infty} = v_{\text{circ}}\sqrt{2\beta - 1}$: 7 km s^{-1} for our fiducial device from 2.7 AU,

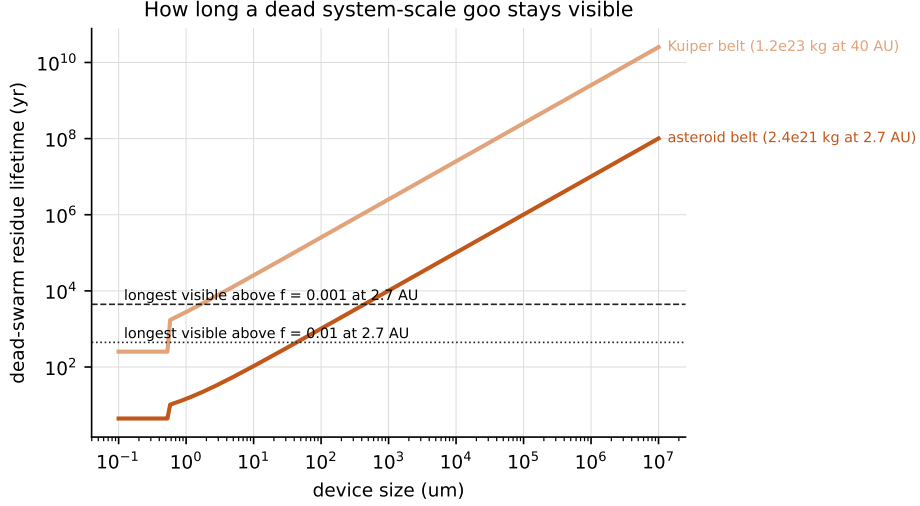


Figure 2: Lifetime of a dead swarm as a function of the size of the bodies it is ground into, for the main-belt mass at 2.7 AU and the Kuiper-belt mass at 40 AU, against the longest a residue can stay above an infrared-excess detection floor (equation 1). Sub-blowout devices leave on an orbit; larger ones on the shorter of the collisional and Poynting–Robertson times.

18 km s^{-1} at $\beta = 1$. A goo that grinds an asteroid belt into micron-scale machines therefore ejects the sub-blowout part of its own mass at stellar speeds, at no cost and without intent. (Devices bound in a swarm also leave whenever a collision frees them; devices on a planet’s surface do not, except as impact ejecta, a channel we return to below.)

The same force governs arrival. An incoming device of $\beta > 1$ is repelled by the target star and can approach no closer than $2(\beta - 1)GM/v_\infty^2$; with the 26 km s^{-1} at which the interstellar medium flows through the heliosphere [Grün et al., 1993, Landgraf, 2000] only $\beta < 1.38$ reaches 1 AU. Devices that both leave and arrive thus occupy a narrow window, $0.21\text{--}0.57 \text{ } \mu\text{m}$ at $\rho = 2 \text{ g cm}^{-3}$; smaller ones are also filtered by the stellar magnetic field at the astropause [Landgraf, 2000], larger ones stay bound to their home star. Once out, the devices join the disc’s velocity field. Epicyclic motion at the $\sim 30 \text{ km s}^{-1}$ dispersion of thin-disc stars confines them to an annulus $\sim 1.6 \text{ kpc}$ wide; differential rotation smears them around it in $(R/\Delta R)$ rotation periods, 1.2 Gyr; radial migration [Sellwood and Binney, 2002, Frankel et al., 2018] carries them across the disc on $\sim 10 \text{ Gyr}$. The passive branch does not have a front. It has a cloud, and the cloud is the annulus.

The belt’s mass in $0.5 \text{ } \mu\text{m}$ devices is 2.3×10^{36} units; in the annulus volume an Earth-sized planet, whose gravitationally focused cross-section at 26 km s^{-1} is 1.2 times its geometric one, sweeps up 200 of them per year, and a biosphere’s mass (10^{15} kg) 8.1×10^{-5} per year. Survival decides everything. Interstellar silicate grains lose their mass to supernova shocks on $\sim 5 \times 10^8 \text{ yr}$ [Jones et al., 1996]; a functioning machine is a more fragile thing than a grain, and we scan its functional e-folding time from 10^6 to 10^9 yr rather than guess it. The cumulative number of intact arrivals on one planet over 5 Gyr (Figure 3) is 1.5×10^5 for a 10^8 yr survival time and 5.9×10^{10} for 10^9 yr , and effectively zero for 10^7 yr because nothing survives the 1.2 Gyr it takes to reach the far side of the annulus. If one intact, general-substrate device suffices to seed a planet, the passive branch seeds every rocky planet in the annulus within a gigayear of a single belt-scale release, provided the devices survive $\gtrsim 10^8 \text{ yr}$ in the interstellar medium.

Two channels we do not compute in detail bracket this one. Impact ejecta from a planetary

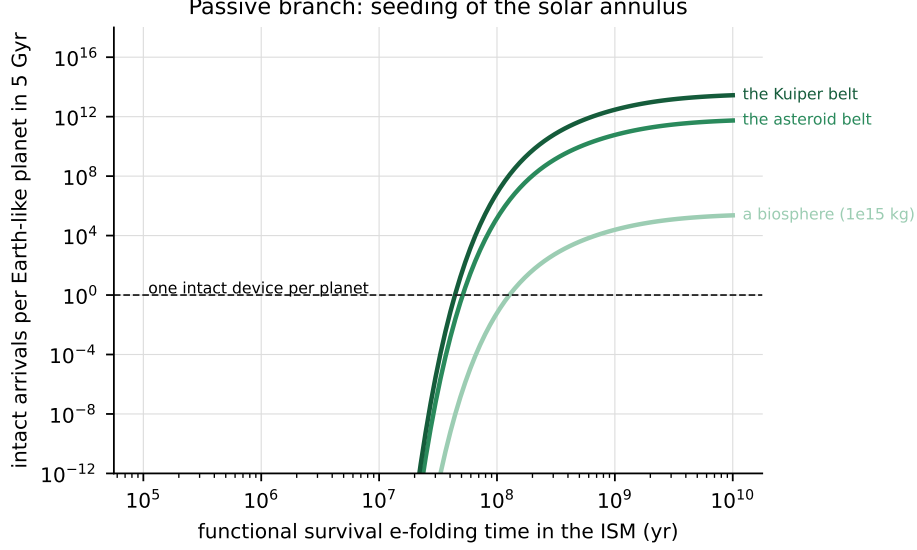


Figure 3: Passive branch: intact devices delivered to one Earth-like planet in the solar annulus over 5 Gyr, after a single release of the mass of a biosphere, the asteroid belt or the Kuiper belt as $0.5 \mu\text{m}$ machines, as a function of their functional survival time in the interstellar medium. Nothing arrives before the annulus is phase-mixed (1.2 Gyr), so survival below $\sim 10^8 \text{ yr}$ delivers nothing.

surface—the lithopanspermia channel [Melosh, 2003, Adams and Spergel, 2005, Worth et al., 2013, Ginsburg et al., 2018]—carries rock-embedded devices at a far lower rate but with far better shielding; the transfer probabilities in that literature apply unchanged to a goo that lives in rock. And a goo whose devices are larger than the blowout size never leaves passively at all: dispersal class is decided at the micron.

6 Between stars, actively: the inherited front

A goo that inherits interstellar propulsion—because it was a probe, or ate a shipyard—is the classical von Neumann front with the purpose removed. For nearest-neighbour hops of length d at ship speed v and a build time τ_b per hop the front moves at $v_f = d/(d/v + \tau_b)$; from the solar circle it fills the disc out to 15 kpc in 504 Myr at Voyager speeds ($10^{-4}c$) with a millennium per hop, 6.4 Myr at $10^{-2}c$ and a century, and 639 kyr at $0.1c$ and a decade (Figure 4). These are the numbers of Hart [1975], Tipler [1980], Newman and Sagan [1981] and everyone since, and they are all short compared to the age of the disc; refinements—percolation with a probability of not launching [Landis, 1998], stellar motions [Carroll-Nellenback et al., 2019], slingshots [Nicholson and Forgan, 2013]—move them by factors of a few. The active branch is the one that makes the Fermi question sharp, and the one the probe literature has always had in mind.

Beyond the Galaxy the reach is bounded by the expansion of the universe. A front launched now at peculiar speed v reaches comoving distance $v \int_{t_0}^{\infty} dt/a(t)$, which for $v = c$ is the cosmic event horizon, 16.6 Gly in Planck-2018 ΛCDM , enclosing $\sim 5 \times 10^9 L^*$ galaxies at 10^{-2} Mpc^{-3} ; at $0.5c$ it is 8.3 Gly and 7×10^8 galaxies, at $0.1c$ 1.7 Gly and 5×10^6 [cf. Armstrong and Sandberg, 2013, Olson, 2015]. Nothing about goo changes these numbers; what goo changes is the prior on whether the front is ever launched, which is the subject of Section 8.

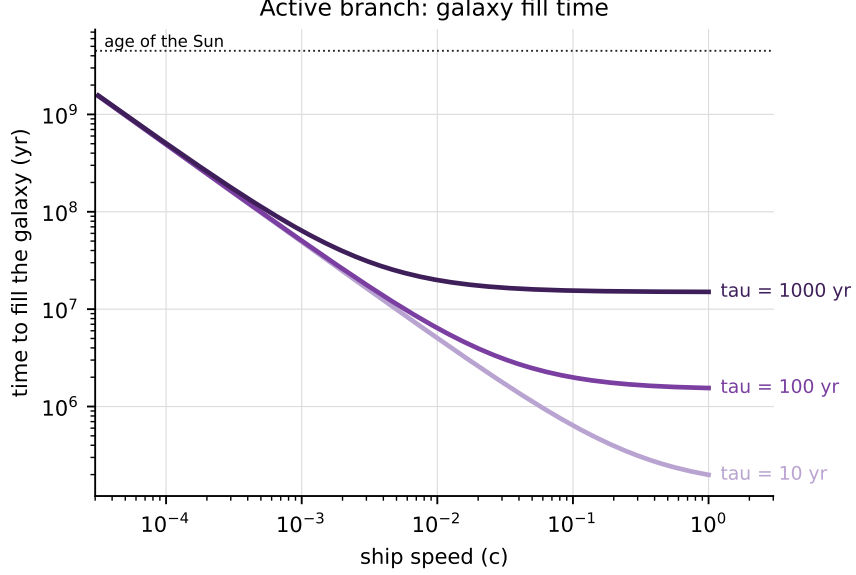


Figure 4: Active branch: time to fill the Galaxy from the solar circle as a function of ship speed for three build times per hop.

7 The observational record

Table 2 collects what has been looked for at each scale, what it would see, and what it found. Three features of the compilation carry the inference.

First, the planetary branch has no observational entry beyond the Earth itself. Second, at the system scale the surveys constrain *live* goo with a memory as long as the goo lives, and *dead* goo with a memory of at most $P_{\text{orb}}/f_{\text{min}}$, so that a null over $N_{\star}$ stars bounds the event rate at $R < 3/(N_{\star}t_{\text{vis}})$: for Hephaistos II’s 5×10^6 stars that is 6 events per year in the whole Galaxy if the residue lasts 10^4 yr, 0.06 if it lasts a megayear, and 6.0×10^{-6} per year— 3×10^4 systems over the 5 Gyr window—if it persists. Third, at the galactic scale the $\hat{G}$ null is both unshadowed and long-memoried: a galaxy-spanning goo that keeps replicating is a galaxy that keeps reprocessing its starlight, and fewer than 3.0×10^{-5} of galaxies do.

The two Solar-System rows are the strongest data we have and the least usable. They say that in 4 Gyr nothing has consumed our belt or our biosphere. But we could not be here to say otherwise, and Section 8 treats that properly.

8 Inference: what the silence constrains

8.1 Model

Over a window $W = 5$ Gyr the Galaxy produced $N \sim \text{Poisson}(\Lambda)$ technological civilisations at uniform times. Each, independently, produced a goo that reaches other stars with probability

$$p = p_{\text{goo}} p_{\text{esc}}, \quad (2)$$

where p_{goo} is the probability that a civilisation suffers a replicator accident at all and p_{esc} the fraction of such accidents that fall in the passive or active interstellar dispersal class. A goo that got out more than t_{fill} ago has reached us. We use $t_{\text{fill}} = 10$ Myr for the active branch and 1.5 Gyr

Table 2: What existing observations say at each scale. “Memory” is how long after the event the signature persists; “shadow” marks whether the datum is subject to observation selection (we could not exist to see the alternative).

scale	observation	result	memory	shadow
planet	thermal emission of rocky exoplanets [Kreidberg et al., 2019, Greene et al., 2023, Zieba et al., 2023]	bare rock on a handful of hot planets; temperate surfaces unmeasurable	—	no
planet (Earth)	continuity of the biosphere for 3.7 Gyr; intact crust	no natural or artificial ecophagy has occurred here	4 Gyr	yes
system (Solar)	intact asteroid and Kuiper belts; no replicator population in the small-body catalogue (this repository’s LOOM and DERELICT channels: 66 686 Rubin orbits, all non-gravitational survivors explained)	no system-scale goo has occurred here	4 Gyr	yes
system (stars)	IRAS [Carrigan, 2009]; Gaia $\times$ WISE waste-heat searches over 5×10^6 stars [Suazo et al., 2024], every candidate since resolved into background galaxies [Ren et al., 2024, Zackrisson et al., 2026]	no live swarm with $f \gtrsim 0.1$ around 5×10^6 stars	live: in-definite; dead: $\leq P_{\text{orb}}/f_{\text{min}}$	no
system (WDs)	this repository’s white-dwarf infrared-excess channel	13 isolated excesses, all $\tau < 0.08$ debris discs; no swarm-like survivor	as above	no
system (old stars)	$12 \mu\text{m}$ excess incidence versus age [Kennedy and Wyatt, 2013]	~ 1 in 10^4 of $>\text{Gyr}$ stars carry bright warm dust; consistent with collisions	$\leq P_{\text{orb}}/f_{\text{min}}$	no
galaxy	$\hat{G}$: WISE over 10^5 galaxies [Wright et al., 2014a, Griffith et al., 2015]; Zackrisson et al. [2015], Garrett [2015]	no galaxy reprocesses $> 85\%$ of its starlight; fraction $< 3.0 \times 10^{-5}$	live: in-definite	no (beyond the reach horizon)

Table 3: 95% upper bounds on p (prior-free exclusion; the log-uniform posterior bounds are within a factor of a few and are given in the released results). “ext” is the 10^5 -galaxy null; “persist” a goo that keeps reprocessing, “dead” one visible for 10^{-4} of the window.

Λ	own galaxy, naive	own galaxy, shadowed	ext, persist	ext, dead
1	3	—	3.0×10^{-5}	0.3
100	0.03	—	3.0×10^{-7}	3.0×10^{-3}
10^4	3.0×10^{-4}	—	3.0×10^{-9}	3.0×10^{-5}

for the passive one, so that the fraction of the window from which an event would have arrived is $f = (W - t_{\text{fill}})/W$.

Three data enter. D_{own} : our system is unconsumed. D_{sys} : no live swarm around $N_{\star} = 5 \times 10^6$ stars. D_{ext} : no persistent reprocessing in $N_{\text{gal}} = 1 \times 10^5$ galaxies. Their likelihoods are

$$\mathcal{L}_{\text{own}} = e^{-\Lambda p f} \quad (\text{naive / self-indication}), \quad \mathcal{L}_{\text{own}} = 1 \quad (\text{anthropic shadow / self-sampling}), \quad (3)$$

$$\mathcal{L}_{\text{ext}} = e^{-N_{\text{gal}} \Lambda p q}, \quad (4)$$

where q is the fraction of the window over which the residue is visible: 1 if the goo keeps replicating, $\sim t_{\text{vis}}/W$ if it dies. The anthropic shadow [Ćirković et al., 2010] is the statement that an observer who exists finds an unconsumed system whatever p is, so that under self-sampling within a fixed reference class D_{own} carries no information; under self-indication, which weights hypotheses by the number of observers they produce, the naive likelihood is recovered because a lower p leaves more systems with observers in them. We report both; the reader’s view of anthropics decides between them, and the point of what follows is that the conclusion about our own risk does not depend on the choice. D_{ext} is not shadowed for any galaxy beyond the reach of an intergalactic front from the Milky Way’s past, which for the passive branch is every galaxy and for the active branch all but the Local Group. We take a log-uniform prior on p over $[10^{-12}, 1]$ for posteriors and also quote the prior-free exclusion $p < -\ln(0.05)/(\Lambda f N q)$ at which the likelihood of the null datum falls to 5%.

8.2 Results

Figure 5 and Table 3 give the 95% bounds. The structure is simple. Our own galaxy’s silence, taken naively, bounds $p \lesssim 0.03$ at $\Lambda = 100$ and 3.0×10^{-4} at 10^4 ; under the shadow it bounds nothing, and the posterior is the prior. The external galaxies rescue the inference: if a galaxy-spanning goo keeps replicating, its absence from 10^5 galaxies gives $p \lesssim 3.0 \times 10^{-7}$ at $\Lambda = 100$, five orders of magnitude below the own-galaxy bound and immune to the shadow; if it dies and leaves a residue visible for 10^{-4} of the window, the bound relaxes to 3.0×10^{-3} . Marginalised over a log-uniform Λ from 10^{-2} to 10^6 —the spread Sandberg et al. [2018] argue the Drake factors actually support—the shadowed own-galaxy bound is 0.25 (the prior), the naive one 0.031, the external persistent one 8.3×10^{-6} and the external dead one 0.011.

8.3 What it means for us

The bounds are on $p = p_{\text{goo}} p_{\text{esc}}$. To say anything about p_{goo} —the quantity the risk literature elicits and the one a civilisation can act on—one must divide by p_{esc} , and this is where the physics of Sections 3–6 bites. The astronomical data multiply the likelihood of the *escaping* branch by a ratio

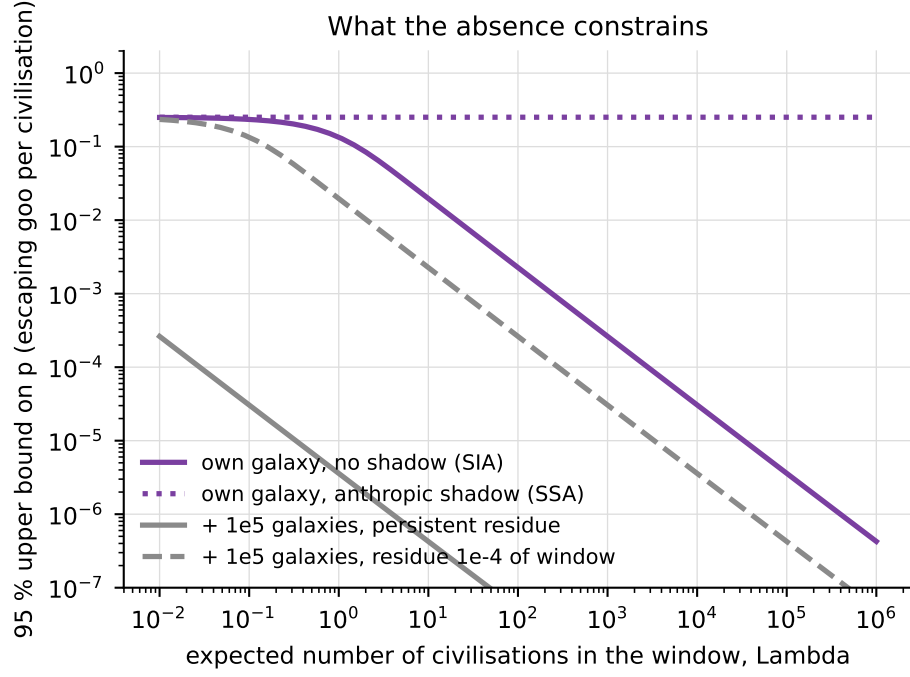


Figure 5: Posterior 95% upper bound on p , the probability per civilisation of a goo that reaches other stars, against the expected number of civilisations Λ in the 5 Gyr window, for the four readings of the data. Under the anthropic shadow our own galaxy's state is uninformative; the external-galaxy null is not shadowed and is the binding constraint whenever the goo's signature persists.

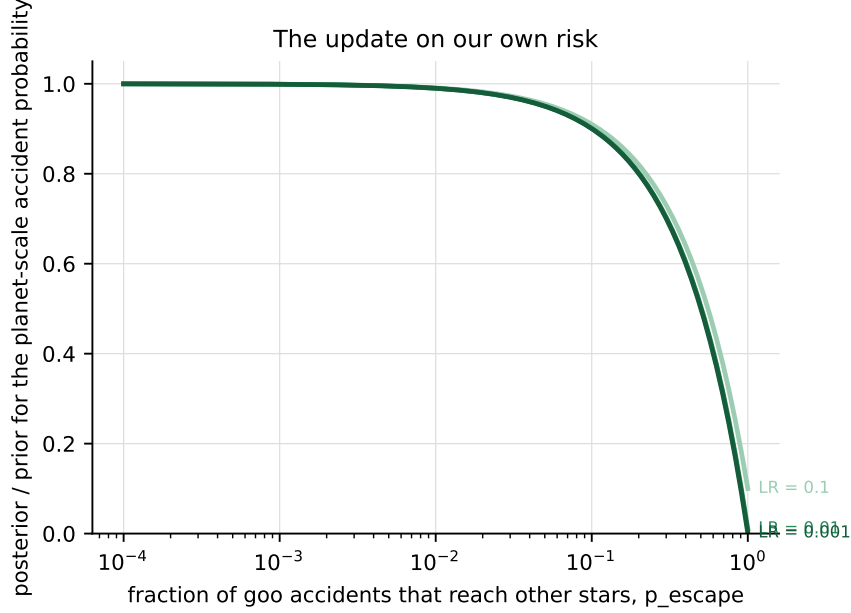


Figure 6: The update on the planet-scale accident probability p_{goo} implied by an astronomical likelihood ratio ℓ on the escaping branch, as a function of the fraction p_{esc} of accidents that reach other stars. The planet-bound branch is untouched; only a civilisation whose accidents would mostly be interstellar learns much from the sky.

$\ell < 1$ and leave the planet-bound branch untouched, so that an elicited prior p_{goo} becomes

$$p'_{\text{goo}} = p_{\text{goo}} \frac{(1 - p_{\text{esc}}) + p_{\text{esc}} \ell}{1 - p_{\text{goo}} p_{\text{esc}} (1 - \ell)}. \quad (5)$$

Figure 6 plots the ratio $p'_{\text{goo}}/p_{\text{goo}}$ for $p_{\text{goo}} = 10^{-3}$. With $\ell = 10^{-3}$ —the external-galaxy null at $\Lambda \sim 100$ against a prior of $p \sim 10^{-3}$ —the update is a factor 0.990 if one per cent of goo accidents reach other stars, 0.90 if a tenth do, and 0.50 if half do. For the elicited priors in the literature—Sandberg and Bostrom [2008] report a median 0.05% for an extinction-level nanotechnology accident by 2100 and 5% for deliberate nanotechnology weapons; Ord [2020] assigns nanotechnology no separate value, treating it within his residual anthropogenic categories at 1 in 30 and 1 in 50 per century—the relevant p_{esc} is the fraction of *our* plausible accidents that would be interstellar. A molecular replicator loose in a laboratory, a factory or a field is planet-bound (Section 3); it becomes system-scale only if released in space or if the civilisation has space industry for it to eat; it becomes interstellar-passive only if it is both in space and below the blowout size, and interstellar-active only if it inherits a starship. For a civilisation at our stage p_{esc} is small, and equation (5) says the astronomical silence changes p_{goo} by a factor within a per cent of unity.

This is the paper’s central conclusion and we state it plainly. *The astronomical record is informative about goo that travels and silent about goo that stays home, and the goo a civilisation at our stage can accidentally make is the kind that stays home.* It should not reassure us about the planetary risk; it does not raise it either. What it does say, with force proportional to Λ , is that a self-replicating machine population that keeps growing across a galaxy is not a common end of civilisations—the external bound is $p < 3.0 \times 10^{-7}$ if a hundred civilisations have arisen per galaxy—and that the classical berserker and runaway-probe scenarios are constrained by the

same data that constrain Kardashev-III engineering. For the passive branch the constraint is on the product of p with the interstellar survival of a machine, and we cannot separate them.

Two corollaries follow for the Great Filter. Hanson [1998] and Bostrom [2008] argue that a filter ahead of us is the pessimistic reading of the silence, and Bostrom that finding independent life nearby would be bad news because it moves the filter forward. Goo is a candidate late filter, and the present calculation says that if it is one, it is an *invisible* one: a filter that consumed civilisations on their own planets would leave a galaxy exactly as quiet as the one we see, and only a filter that spilled out would have announced itself. The silence therefore does not distinguish a planet-bound goo filter from no filter; it distinguishes only against goo filters that travel. Conversely, Ćirković [2004] noted that catastrophic explanations of the silence have an optimistic side—they are consistent with life being common—and a planet-bound goo filter has that property in full.

9 The other replicators

The argument does not depend on the replicator being molecular. It depends on three things: that the population grows without external limit, that its reach is set by dispersal, and that its astronomical visibility is set by whether it keeps metabolising where a survey can see. Table 4 runs the other self-replicating technologies with which a civilisation could end itself through the same three questions. Engineered organisms [Millett and Snyder-Beattie, 2017, Esvelt, 2022] and mirror life [Adamala et al., 2024] are planet-bound by construction: their dispersal is the biosphere’s, their reach the biosphere’s, and their astronomical signature at most the loss of a biosignature. Gene drives [Noble et al., 2018] are the same with a smaller substrate. Macroscopic self-replicating machines—the lunar factory of Freitas and Gilbreath [1982], the “clanking replicators” of Moses and Chirikjian [2020], the space-industry bootstraps of Metzger et al. [2013]—are system-scale by construction and are the branch the waste-heat surveys already bound; their units are too large for passive escape and their reach stops at the system unless they inherit a ship. Computational replicators—network worms [Moore et al., 2003, Staniford et al., 2002] and, latterly, self-replicating AI agents [Pan et al., 2024]—have the fastest doubling times of any replicator ever built (the Slammer worm doubled every 8.5s and infected 75 000 hosts in ten minutes) and the smallest reach: the substrate is the network, and the network stops at the edge of the civilisation’s own hardware. Their astronomical signature is nil unless they capture the physical economy, at which point they are one of the other rows. Only the deliberately interstellar replicator—the probe—has a reach the sky can see, and it is the one replicator whose safety a civilisation designs on purpose.

The general lesson is the same as the specific one. Accidental replicators inherit the dispersal of the environment they escape into, and for a civilisation at our stage that environment is a planet. The astronomical silence is therefore a statement about the one class of replicator we would have to build on purpose, and about the civilisations that built it. A companion paper develops this generalisation and its policy content.

10 Discussion

Assumptions that matter. The calculation assumes substrate generality: a device that eats an asteroid can eat a planet’s surface, and one intact seed suffices. Nothing we know of can do this, and Phoenix and Drexler [2004] and Freitas [2000] argue that nothing easily could; the paper’s numbers are conditional on the accident having happened. It assumes the error catastrophe does not end the goo before the scale in question is reached. Error-catastrophe authors [2016] argue that finite-fidelity copying makes an optimal probe design fail on a bounded number of generations,

Table 4: The other self-replicating technologies a civilisation could die of, by the three properties that govern reach and visibility.

replicator		doubling	dispersal	reach	astronomical signature
molecular assembler	assembler	10^2 – 10^4 s	none / space / blowout / ship	planet / system / annulus / galaxy	none / waste heat / seeding / front
engineered organism, mirror life	organism	hours	biosphere	planet	loss of a biosignature
gene drive		generations	population	planet	none
macroscopic machine, space industry	machine	months–years	space transport	system	waste heat at the feedstock’s orbit
network agent	worm, AI	seconds–hours	network	the civilisation’s hardware	none, unless it takes the economy
von Neumann probe	Neumann	years	starship	galaxy, cosmos	front; Kardashev-III reprocessing

and the active branch is the most exposed: 10^4 hops across the Galaxy is 10^4 generations at least. A goo that dies of copying errors before it crosses the Galaxy is, for our purposes, a system-scale goo, and the inference moves it to that row. It assumes no competition: Lotka–Volterra probe authors [2022] show that predator–prey dynamics between replicator strains can hold a population far below the feedstock limit. And the passive branch assumes a survival time that no one has measured for anything but grains.

Live versus dead. The single most consequential result is equation (1). It means that a goo that stops is invisible on any timescale a survey samples, and that what the waste-heat surveys bound is a goo that never stops. Whether a real goo would stop—exhausting its feedstock, losing to its own mutants, or grinding into fragments it can no longer use—is the question on which the system-scale bound turns, and it is a question about the design of replicators, not about astronomy.

The anthropic choice. We have shown both readings of our own system’s silence. We note that the choice does not change the conclusion in Section 8.3, because that conclusion rests on the external galaxies, which no reading shadows. It does change what one may say about the passive branch, whose gigayear timescale means the external galaxies are unaffected by it (an annulus in another galaxy is no less consumed for being far away) but whose signature is not a galaxy-wide reprocessing: a passively seeded annulus is a set of consumed systems, each of which is visible only while alive. The passive branch is therefore bounded by the system-scale surveys, not by $\hat{G}$, and the bound is weaker by the ratio of the survey volume to the Galaxy.

What would change the answer. Three observations would. A temperate rocky planet with a refined, uniform surface and no atmosphere chemistry—the planetary residue—is beyond current instruments but not beyond a large direct-imaging telescope; a single such detection would be the first entry in the planetary row. A debris disc around an old, metal-poor or halo star, where no natural reservoir can make one (this repository’s OSSUARY channel; Lacki 2025 named the population), would be the system-scale residue in the one place its natural background is absent. And a swarm whose temperature matches its host’s belt rather than its habitable zone—waste heat where the rock was, not where the energy is wanted—would be the discriminant between goo and purpose in any future Dyson candidate.

11 Conclusions

1. No published work computes the propagation of an uncontrolled self-replicator beyond its planet, compiles the observational constraints on it, or draws the implication for the civilisation asking. This paper does all three.
2. Replication rate is almost never the bottleneck. In place, a planet’s crust is a 4.5 kyr project and its bulk a 1.1 Myr one at any doubling time, because the surface cannot shed the heat faster; a planetary system’s small bodies are transport-limited to 14 yr–1.1 kyr; the Galaxy is a 639 kyr–504 Myr project for an active front and a 1.2 Gyr one for a passive cloud. Dispersal class, decided at the micron and at the shipyard, sets the reach.
3. A detectable dead residue is a short-lived one: at most $P_{\text{orb}}/f_{\text{min}}$ above the detection floor. The waste-heat surveys bound live goo; dead goo is invisible on any timescale surveyed.
4. A replicator smaller than 0.57 μm leaves its system on starlight alone, and the 0.21–0.57 μm class reaches the next habitable zone; a single belt-scale release seeds every rocky planet in the solar annulus within a gigayear if the devices survive $\gtrsim 10^8$ yr in the interstellar medium. Passive goo has no front; it has a cloud.
5. The observational record constrains, per civilisation, $p \lesssim 3.0 \times 10^{-7}$ for a persistent galaxy-spanning goo at $\Lambda = 100$, from the 10^5 -galaxy null, which is neither anthropically shadowed nor short-memoried; our own system’s silence constrains nothing under the shadow and $p \lesssim 0.03$ without it.
6. Mapped onto the planet-scale accident probability that a civilisation at our stage can act on, the silence changes it by a factor 0.990 if one per cent of accidents would be interstellar. The sky is informative about goo that travels and silent about goo that stays home; ours would stay home.

Data and code availability

The model, configuration, tests and figure code are in the `seti.goo` package of the accompanying repository; `python -m seti.goo.run --stage all` regenerates every number in this paper from `config/goo.yaml`. The compiled observational record cites public surveys; the in-house channel results referenced in Table 2 are committed under `results/` in the same repository.

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[To be added.]

References

- Katarzyna P. Adamala, Deepa Agashe, Yasmine Belkaid, et al. Confronting risks of mirror life. *Science*, 386:1351–1353, 2024. doi: 10.1126/science.ads9158.
- Fred C. Adams and David N. Spergel. Lithopanspermia in star-forming clusters. *Astrobiology*, 5: 497–514, 2005. doi: 10.1089/ast.2005.5.497.

- Stuart Armstrong and Anders Sandberg. Eternity in six hours: Intergalactic spreading of intelligent life and sharpening the Fermi paradox. *Acta Astronautica*, 89:1–13, 2013. doi: 10.1016/j.actaastro.2013.04.002.
- Yinon M. Bar-On, Rob Phillips, and Ron Milo. The biomass distribution on Earth. *PNAS*, 115: 6506–6511, 2018. doi: 10.1073/pnas.1711842115.
- Rasmus Bjørk. Exploring the Galaxy using space probes. *International Journal of Astrobiology*, 6: 89–93, 2007. doi: 10.1017/S1473550407003709.
- Nick Bostrom. Existential risks: Analyzing human extinction scenarios and related hazards. *Journal of Evolution and Technology*, 9, 2002.
- Nick Bostrom. Where are they? Why I hope the search for extraterrestrial life finds nothing. *MIT Technology Review*, 2008. asserted: magazine, May/June issue.
- Joseph A. Burns, Philippe L. Lamy, and Steven Soter. Radiation forces on small particles in the solar system. *Icarus*, 40:1–48, 1979. doi: 10.1016/0019-1035(79)90050-2.
- Jr. Carrigan, Richard A. IRAS-based whole-sky upper limit on Dyson spheres. *ApJ*, 698:2075–2086, 2009. doi: 10.1088/0004-637X/698/2/2075.
- Jonathan Carroll-Nellenback, Adam Frank, Jason Wright, and Caleb Scharf. The Fermi paradox and the Aurora effect: Exo-civilization settlement, expansion, and steady states. *AJ*, 158:117, 2019. doi: 10.3847/1538-3881/ab31a3.
- Milan M. Ćirković. Cosmic irony: SETI optimism from catastrophes? *Contact in Context*, 2, 2004.
- Milan M. Ćirković, Anders Sandberg, and Nick Bostrom. Anthropic shadow: Observation selection effects and human extinction risks. *Risk Analysis*, 30:1495–1506, 2010. doi: 10.1111/j.1539-6924.2010.01460.x.
- Carlos Cotta and Álvaro Morales. A computational analysis of galactic exploration with space probes: implications for the Fermi paradox. *JBIS*, 62:82–88, 2009.
- K. Eric Drexler. *Engines of Creation: The Coming Era of Nanotechnology*. Anchor/Doubleday, New York, 1986. asserted: book.
- Manfred Eigen. Selforganization of matter and the evolution of biological macromolecules. *Naturwissenschaften*, 58:465–523, 1971. doi: 10.1007/BF00623322.
- Alex Ellery. Self-replicating probes are imminent – implications for SETI. *International Journal of Astrobiology*, 21:212–242, 2022a. doi: 10.1017/S1473550422000106.
- Alex Ellery. Curbing the fruitfulness of self-replicating machines. *International Journal of Astrobiology*, 21:243–259, 2022b. doi: 10.1017/S1473550422000155.
- Alex Ellery. Technosignatures of self-replicating probes in the solar system. *arXiv e-prints*, 2025.
- Error-catastrophe authors. Why is there no von neumann probe on ceres? error catastrophe can explain the fermi-hart paradox. *arXiv e-prints*, 2016. author list to be taken from results/goolit/verification.json.

- Kevin M. Esvelt. Delay, detect, defend: Preparing for a future in which thousands can release new pandemics. Technical report, Geneva Centre for Security Policy, 2022. asserted: report.
- Neige Frankel, Hans-Walter Rix, Yuan-Sen Ting, Melissa Ness, and David W. Hogg. Measuring radial orbit migration in the Galactic disk. *ApJ*, 865:96, 2018. doi: 10.3847/1538-4357/aadba5.
- Jr. Freitas, Robert A. A self-reproducing interstellar probe. *JBIS*, 33:251–264, 1980.
- Jr. Freitas, Robert A. Some limits to global ecophagy by biovorous nanoreplicators, with public policy recommendations. Technical report, Foresight Institute, 2000. asserted: report.
- Jr. Freitas, Robert A. and William P. Gilbreath. Advanced automation for space missions. Technical Report CP-2255, NASA, 1982. asserted: report.
- M. A. Garrett. Application of the mid-IR radio correlation to the $\hat{g}$ sample and the search for advanced extraterrestrial civilisations. *A&A*, 581:L5, 2015. doi: 10.1051/0004-6361/201526687.
- Idan Ginsburg, Manasvi Lingam, and Abraham Loeb. Galactic panspermia. *ApJL*, 868:L12, 2018. doi: 10.3847/2041-8213/aaef2d.
- Thomas P. Greene, Taylor J. Bell, Elsa Ducrot, et al. Thermal emission from the Earth-sized exoplanet TRAPPIST-1 b using JWST. *Nature*, 618:39–42, 2023. doi: 10.1038/s41586-023-05951-7.
- Roger L. Griffith, Jason T. Wright, Jessica Maldonado, Matthew S. Povich, Steinn Sigurdsson, and Brendan Mullan. The $\hat{g}$ infrared search for extraterrestrial civilizations with large energy supplies. III. the reddest extended sources in WISE. *ApJS*, 217:25, 2015. doi: 10.1088/0067-0049/217/2/25.
- E. Grün, H. A. Zook, M. Baguhl, et al. Discovery of Jovian dust streams and interstellar grains by the Ulysses spacecraft. *Nature*, 362:428–430, 1993. doi: 10.1038/362428a0.
- Robin Hanson. The great filter: are we almost past it?, 1998. asserted: online essay.
- Robin Hanson, Daniel Martin, Calvin McKay, and Jonathan Wehowsky. If loud aliens explain human earliness, quiet aliens are also rare. *ApJ*, 922:182, 2021. doi: 10.3847/1538-4357/ac2369.
- Michael H. Hart. An explanation for the absence of extraterrestrials on earth. *QJRAS*, 16:128–135, 1975.
- A. P. Jones, A. G. G. M. Tielens, and D. J. Hollenbach. Grain shattering in shocks: The interstellar grain size distribution. *ApJ*, 469:740, 1996. doi: 10.1086/177823.
- G. M. Kennedy and M. C. Wyatt. The bright end of the exo-Zodi luminosity function: disc evolution and implications for exo-Earth detectability. *MNRAS*, 433:2334–2356, 2013. doi: 10.1093/mnras/stt900.
- Laura Kreidberg, Daniel D. B. Koll, Caroline Morley, et al. Absence of a thick atmosphere on the terrestrial exoplanet LHS 3844b. *Nature*, 573:87–90, 2019. doi: 10.1038/s41586-019-1497-4.
- Brian C. Lacki. Ground to dust: Collisional cascades and the fate of Kardashev II megaswarms. *ApJ*, 985:191, 2025.
- Markus Landgraf. Modeling the motion and distribution of interstellar dust inside the heliosphere. *Journal of Geophysical Research*, 105:10303–10316, 2000. doi: 10.1029/1999JA900243.

- Geoffrey A. Landis. The Fermi paradox: an approach based on percolation theory. *JBIS*, 51: 163–166, 1998.
- Lotka–Volterra probe authors. Lotka–volterra models for extraterrestrial self-replicating probes. *arXiv e-prints*, 2022. author list to be taken from results/goolit/verification.json.
- H. J. Melosh. Exchange of meteorites (and impact ejecta) between stellar systems. *Astrobiology*, 3:207–215, 2003. doi: 10.1089/153110703321632525.
- Philip T. Metzger, Anthony Muscatello, Robert P. Mueller, and James Mantovani. Affordable, rapid bootstrapping of the space industry and solar system civilization. *Journal of Aerospace Engineering*, 26:18–29, 2013. doi: 10.1061/(ASCE)AS.1943-5525.0000236.
- Piers Millett and Andrew Snyder-Beattie. Existential risk and cost-effective biosecurity. *Health Security*, 15:373–383, 2017. doi: 10.1089/hs.2017.0028.
- David Moore, Vern Paxson, Stefan Savage, Colleen Shannon, Stuart Staniford, and Nicholas Weaver. Inside the Slammer worm. *IEEE Security & Privacy*, 1(4):33–39, 2003. doi: 10.1109/MSECP.2003.1219056.
- Matthew S. Moses and Gregory S. Chirikjian. Robotic self-replication. *Annual Review of Control, Robotics, and Autonomous Systems*, 3:1–24, 2020. doi: 10.1146/annurev-control-071819-010010.
- William I. Newman and Carl Sagan. Galactic civilizations: Population dynamics and interstellar diffusion. *Icarus*, 46:293–327, 1981. doi: 10.1016/0019-1035(81)90135-4.
- Arwen Nicholson and Duncan Forgan. Slingshot dynamics for self-replicating probes and the effect on exploration timescales. *International Journal of Astrobiology*, 12:337–344, 2013. doi: 10.1017/S1473550413000244.
- Charleston Noble, Ben Adlam, George M. Church, Kevin M. Esvelt, and Martin A. Nowak. Current CRISPR gene drive systems are likely to be highly invasive in wild populations. *eLife*, 7:e33423, 2018. doi: 10.7554/eLife.33423.
- S. Jay Olson. Homogeneous cosmology with aggressively expanding civilizations. *Classical and Quantum Gravity*, 32:215025, 2015. doi: 10.1088/0264-9381/32/21/215025.
- Toby Ord. *The Precipice: Existential Risk and the Future of Humanity*. Hachette, 2020. asserted: book.
- Xudong Pan, Jiarun Dai, Yihe Fan, and Min Yang. Frontier AI systems have surpassed the self-replicating red line. *arXiv e-prints*, 2024.
- Chris Phoenix and Eric Drexler. Safe exponential manufacturing. *Nanotechnology*, 15:869–872, 2004. doi: 10.1088/0957-4484/15/8/001.
- Martin Rees. *Our Final Hour*. Basic Books, 2003. asserted: book.
- Tongtian Ren, Michael A. Garrett, and Andrew P. V. Siemion. Background contamination of the Project Hephaistos Dyson spheres candidates. *Research Notes of the AAS*, 8:145, 2024. doi: 10.3847/2515-5172/ad5017.
- Carl Sagan and William I. Newman. The solipsist approach to extraterrestrial intelligence. *QJRAS*, 24:113–121, 1983.

- Anders Sandberg and Nick Bostrom. Global catastrophic risks survey. Technical Report Technical Report 2008-1, Future of Humanity Institute, Oxford University, 2008. asserted: report.
- Anders Sandberg, Eric Drexler, and Toby Ord. Dissolving the Fermi paradox. *arXiv e-prints*, 2018.
- J. A. Sellwood and J. J. Binney. Radial mixing in galactic discs. *MNRAS*, 336:785–796, 2002. doi: 10.1046/j.1365-8711.2002.05806.x.
- Stuart Staniford, Vern Paxson, and Nicholas Weaver. How to Own the Internet in your spare time. In *Proceedings of the 11th USENIX Security Symposium*, 2002. asserted: proceedings.
- Adam Stevens, Duncan Forgan, and Jack O’Malley James. Observational signatures of self-destructive civilizations. *International Journal of Astrobiology*, 15:333–344, 2016. doi: 10.1017/S1473550415000397.
- Matías Suazo, Erik Zackrisson, Jason T. Wright, Andreas J. Korn, and Macy Huston. Project Hephaistos – I. upper limits on partial Dyson spheres in the Milky Way. *MNRAS*, 512:2988–3000, 2022. doi: 10.1093/mnras/stac280.
- Matías Suazo, Erik Zackrisson, Priyatam K. Mahto, Fabian Lundell, Carl Nettelblad, Andreas J. Korn, Jason T. Wright, and Suman Majumdar. Project Hephaistos – II. Dyson sphere candidates from Gaia DR3, 2MASS, and WISE. *MNRAS*, 531:695–707, 2024. doi: 10.1093/mnras/stae1186.
- Frank J. Tipler. Extraterrestrial intelligent beings do not exist. *QJRAS*, 21:267–281, 1980.
- Keith B. Wiley. The Fermi paradox, self-replicating probes, and the interstellar transportation bandwidth. *arXiv e-prints*, 2011.
- R. J. Worth, Steinn Sigurdsson, and Christopher H. House. Seeding life on the moons of the outer planets via lithopanspermia. *Astrobiology*, 13:1155–1165, 2013. doi: 10.1089/ast.2013.1028.
- Jason T. Wright, Roger L. Griffith, Steinn Sigurdsson, Matthew S. Povich, and Brendan Mullan. The $\hat{g}$ infrared search for extraterrestrial civilizations with large energy supplies. II. framework, strategy, and first result. *ApJ*, 792:27, 2014a. doi: 10.1088/0004-637X/792/1/27.
- Jason T. Wright, Brendan Mullan, Steinn Sigurdsson, and Matthew S. Povich. The $\hat{g}$ infrared search for extraterrestrial civilizations with large energy supplies. I. background and justification. *ApJ*, 792:26, 2014b. doi: 10.1088/0004-637X/792/1/26.
- Erik Zackrisson, Per Calissendorff, Saghar Asadi, and Anders Nyholm. Extragalactic SETI: The Tully–Fisher relation as a probe of Dysonian astroengineering in disk galaxies. *ApJ*, 810:23, 2015. doi: 10.1088/0004-637X/810/1/23.
- Erik Zackrisson, Arjan Bik, M. Ali Asgar, S. Curtis, Jason T. Wright, Tongtian Ren, et al. Project Hephaistos – IV. James Webb Space Telescope observations of two Dyson sphere candidates. *arXiv e-prints*, 2026.
- Sebastian Zieba, Laura Kreidberg, Elsa Ducrot, et al. No thick carbon dioxide atmosphere on the rocky exoplanet TRAPPIST-1 c. *Nature*, 620:746–749, 2023. doi: 10.1038/s41586-023-06232-z.